

PROBLEM ANALYSIS

In this task, we are dealing with two images: the inner image and the outer image. Both images initially have different rotations, resulting in misalignment. The goal is to calculate the rotation angle required to align the two images so they form a complete, coherent image (similar to a puzzle). The images need to be rotated until their structures match perfectly.

Key Observations:

- Misalignment: The inner and outer images are not aligned in their initial state, meaning their features don't match at angle 0° .
- Puzzle Analogy: The images are like two puzzle pieces — they only fit correctly when rotated to the correct position.

Rotation Methods:

There are three possible ways to solve the alignment problem:

1. Rotate Only the Outer Image: Rotate only the outer image until it aligns with the inner image. This method focuses on adjusting the outer image's position relative to the inner one.
2. Rotate Only the Inner Image: Rotate only the inner image to align with the outer image. This is simpler but may not always be the best choice for certain image pairs.
3. Rotate Both Images Simultaneously: This is the most similar to the puzzle-like scroll, where both the inner and outer images are rotated simultaneously but in opposite directions (as in the scroll movement where each point of adjustment rotates both images by 1 degree).

Approach:

Given the input as a pair of inner and outer images, the algorithm must determine the optimal rotation angle that results in correct alignment. Although simultaneous rotation best reflects the physical puzzle interaction, from a computational standpoint, rotating just one image is mathematically equivalent and simpler to implement. The algorithm will:

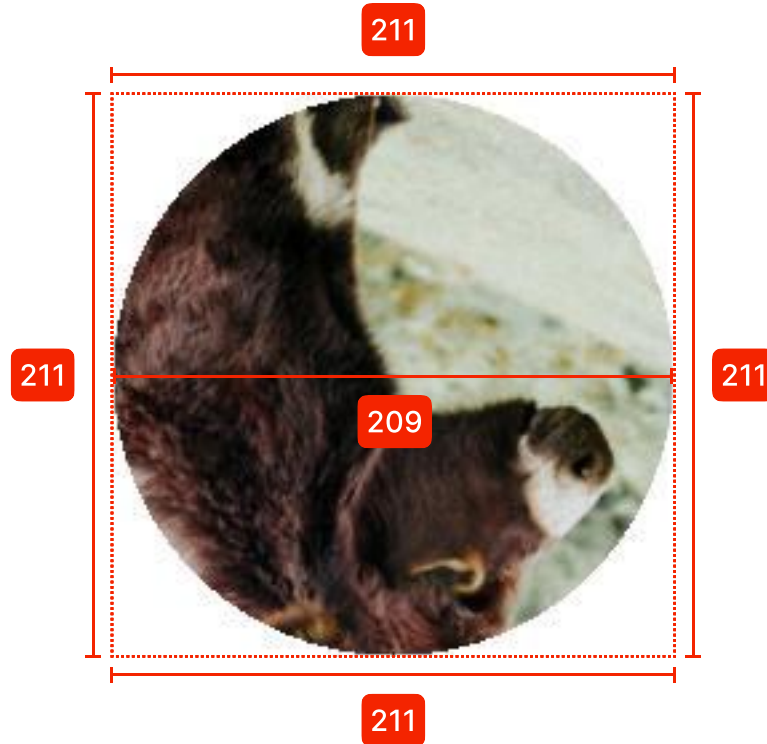
- Analyze the features of both images.
- Evaluate candidate rotations (e.g., via brute force search or feature matching) to determine the angle that maximizes alignment.
- Return the optimal rotation angle.

This method should generalize to any image pair, regardless of their default rotations or specific features, ensuring that the inner image fits perfectly into the outer donut.

MEASUREMENT ANALYSIS

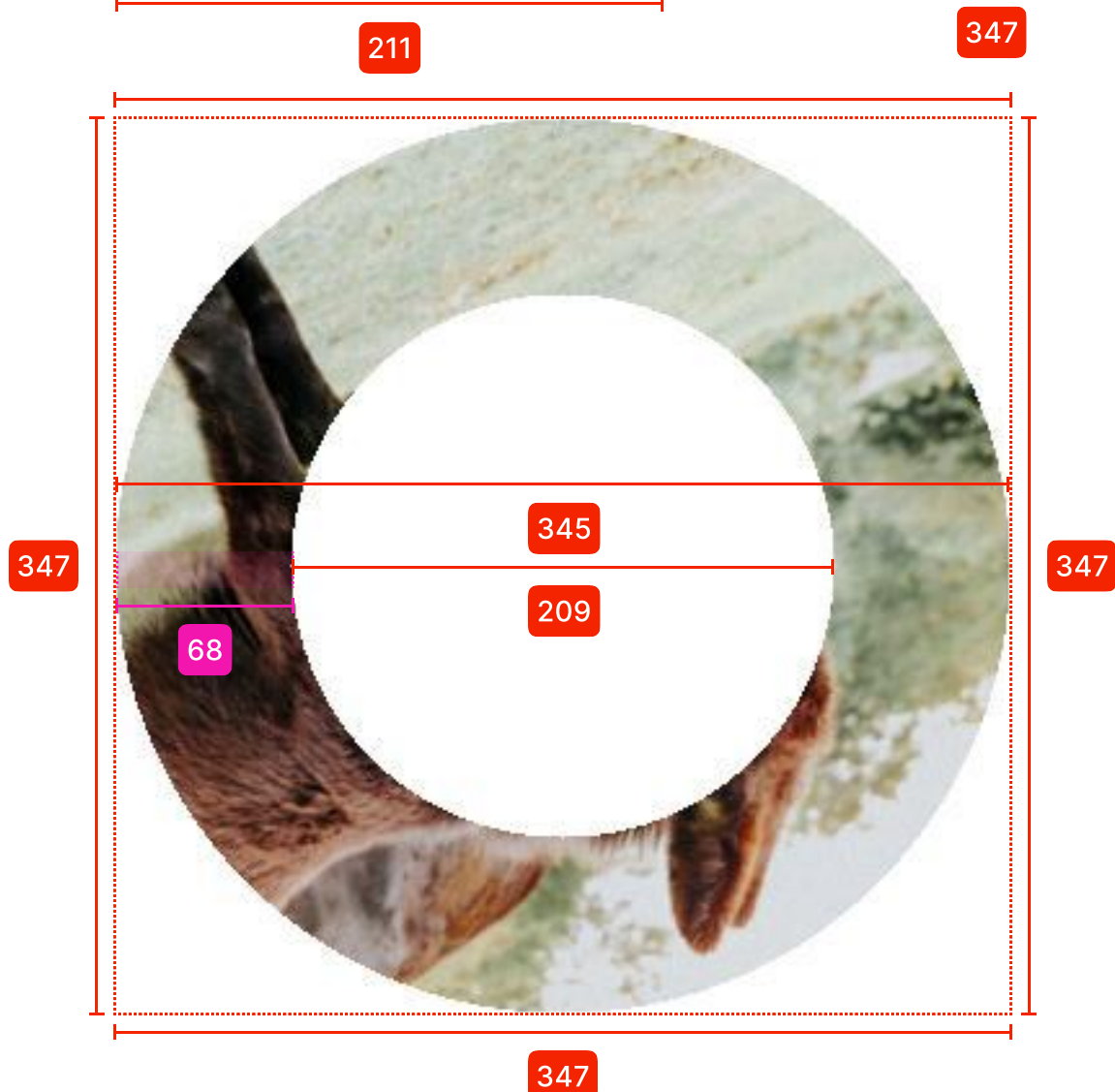
Inner Image:

- Total Size: 211 px by 211 px.
- Area of Interest: Only the 209 px diameter circle is relevant.



Outer Image:

- Total Size: 347 px by 347 px.
- Donut Border: The donut's border is 68 px thick.
- Hole Size: The hole in the donut has a 209 px diameter, which matches the inner circle.



The 209 px circle from the inner image must be aligned within the 209 px hole of the outer image.

PROBLEM EXAMPLE

Default

Match

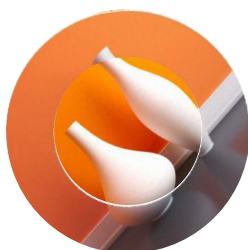
Rotation Angle



-49.85°



142.48°



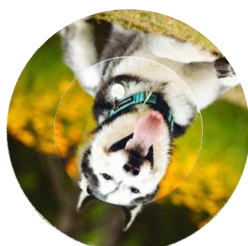
97.67°



-134.29°



92.45°



91.65